MILITARY SPECIFICATION

CONTROL PANEL MARK 264 MOD 1

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 This specification establishes the requirements for the procurement of Control Panel Mark 264 Mod 1, BUWEPS LD 162136, which presets and controls the firing of Torpedoes Mark 44 all Mods and Mark 46 Mod 0, launched from Torpedo Tube Mark 32 Mods 0, 1, and 2.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

MILITARY

MIL-P-116	Preservation, Methods of
MIL-A-5092	Adhesive, Rubber (Synthetic and Reclaimed Rubber Base)
MIL-G-12803	Gasket Materials, Non-metallic
MIL-I-16910	Interference Measurement, Radio, Methods and Limits; 14 Kilocycles to 1000 Megacycles

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MIL-I-26860	Indicator, Humidity, Plug, Color Change

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

NAVY DEPARTMENT

BUWEPS LD 162136	Control Panel Mark 264 Mod 1

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

NAVWEPS QAP 609	Quality Assurance Provisions for Control Panel Mark 264 Mod 1

(Copies of specifications, standards, and drawings requested by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATION

Handbook H28	Screw Thread Standards for the Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

UNIFORM CLASSIFICATION COMMITTEE

UNIFORM FREIGHT CLASSIFICATION RULES

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(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois. Orders for the above publication should cite the latest edition and supplements thereto.)

AMERICAN TRUCKING ASSOCIATION, INC., PUBLICATION

MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1424 16th St., N. W., Washington 6, D. C.)

3. REQUIREMENTS

3.1 Description. - The Control Panel Mark 264 Mod 1 is a torpedo presetting and firing device, housed in a cast aluminum case measuring 18 7/8 by 13 3/4 by 14 1/2 inches. Functionally, the control panel selects one of six torpedoes, Mark 44 all Mods or Mark 46 Mod 0, stowed in the barrels of two Torpedo Tubes Mark 32 Mods 0, 1, and 2, and transmits signals for presetting Initial Search Depth (ISD) and Search Floor or Gyro Angle into the selected torpedo. The control panel also provides a standby-firing switch for controlling the firing of the torpedo, and mechanical aids which indicate the type of torpedo stowed in each of the six barrels and the allowable settings for each torpedo.

3.2 General Requirements.

3.2.1 Pre-production Samples. - Unless otherwise specified in the contract or order, pre-production samples of the control panel shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the contractor, prior to his starting production, is capable of producing items that comply with the technical requirements of the contract. Provision of samples shall be in accordance with 4.1.2.

3.2.2 Materials. - Unless otherwise specified in the contract or order, all materials and purchased parts entering into the final assembly of the control panel shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BUWEPS LD 162136.

3.2.3 Finishes, Coatings, Construction and Dimensions. - Finishes, coatings, construction and dimensions for the control panel shall be strictly in accordance with the applicable drawings, specifications, and publications listed in BUWEPS LD 162136.

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3.2.4 Moisture, Fungus and Corrosion. - The control panel shall be rendered moisture, fungus and corrosion resistant in accordance with detail requirements specified in the drawings and specifications listed in BUWEPS LD 162136.

3.2.5 Marking. - Marking of assemblies, sub-assemblies and parts shall be in accordance with the applicable drawing listed in BUWEPS LD 162136.

3.2.6 Interchangeability. - Component parts of assembled units fabricated in strict agreement with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of the unit, selection or fitting is required, then changes in dimensions are necessary and revision and correction of applicable drawings are required. (See 3.2.9)

3.2.7 Threads. - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class of fit for threads shall be specified on the applicable drawings, and the NOT-GO gaging practice shall be as set forth in the aforementioned Handbook H28.

3.2.8 Requests for Deviation. - Each request that a deviation be authorized shall have been submitted to the Procuring Activity, and the deviation shall have been authorized by the Bureau of Naval Weapons or by a Naval Activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation on the item. The deviation shall apply only to this specific item or operation. The request shall define the suggested deviation in sufficient detail to permit the Procuring Activity to evaluate its effect on performance and reliability and the reasons for the suggested deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the extent of the departure, the identifying number of the item, the contractor (or bidder) and the contract number. A deviation is essentially temporary in character, and usually represents a departure limited to a specific period, to a specific number of items, or to the contract to which it applies. It may require an equitable reduction in contract price if a decrease in the cost of performance results therefrom.

3.2.9 Conflicting Requirements. - Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.3 Performance Requirements and Product Characteristics. - Control Panel Mk 264 Mod 1, prior to, during, and after being subjected to the environmental tests outlined in Section 4, shall meet the following performance requirements and product characteristics.

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3.3.1 Operation. - The control panel shall be capable of inserting initial search depth and search floor or gyro angle settings into a torpedo and obtain a set indication for each and every setting of the initial search depth and search floor/gyro angle selector switches.

3.3.2 Standby-firing Circuit. - The control panel shall be capable of transmitting signals of sufficient voltage and frequency to cause detonation of the torpedo squib and firing of the torpedo itself, in that order.

3.3.2.1 Time Delay. - The time lapse between the squib detonating signal and torpedo firing signal shall not be less than 92 milliseconds.

3.3.3 Circuit Breakers. - The control panel shall provide short circuit protection to the AC and DC outputs.

3.3.4 Power Supply. - The control panel shall operate satisfactorily when furnished with the electrical supplies listed in table I.

Table I. Power Supplies

Nominal Voltage (volts)	Frequency (c.p.s.)	Phase	Power (watts)
115 ± 5%	60	1	850
28 ± 1 V	D.C.	-	150

3.3.5 Radio Frequency and Low Frequency Noise Interference. - The maximum radiated and conducted noise interference and electrical disturbances shall not exceed the following values:

Frequency Range	Limit Microvolts (Quasi Peak)
15 KC to 59 KC	5000
60 KC to 99 KC	1000
100 KC to 149 KC	100
150 KC to 4 MC	50
5 MC to 400 MC	5

3.3.6 Dielectric. - Control panel circuits with operating potentials between 28 and 125 volts shall be capable of withstanding a potential of 1000 volts AC; circuits with operating potentials not over 28 volts shall be capable of withstanding a potential of 300 volts AC.

3.3.7 Insulation Resistance. - Each circuit of the control panel shall show an insulation resistance of at least three megohms at a potential of 500 volts DC between terminal and ground.

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3.3.8 Operating Life. - The control panel shall be capable of withstanding the operating life test specified in 4.5.8.

3.4 Environmental Requirements.

3.4.1 Vibration. - Unless otherwise specified in the contract or order, the control panel, shall be capable of withstanding vibration requirements as follows:

3.4.1.1 Vibration, Transportation, Storage, and Handling. - The control panel, packed and packaged as defined in 5 and mounted rigidly as specified in 4.5.1.1.1, shall be capable of withstanding the tests as specified in 4.5.1.1.2 and thereafter, shall show no damage and shall meet all the performance and product requirements of this specification.

3.4.1.2 Vibration, Operating Condition. - The control panel, self-mounted as specified in 4.5.1.2.1 shall be capable of withstanding vibration tests as specified in 4.5.1.2.2 and thereafter shall show no damage and meet all performance requirements and product characteristics of this specification.

3.4.2 Shock. - Unless otherwise specified in the contract or order, the control panel shall be capable of withstanding shock tests as follows:

3.4.2.1 Shock, Transportation, Storage, and Handling. - The control panel packed and packaged as defined in 5 shall be capable of withstanding tests as specified in 4.5.2.1 and thereafter show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.2.2 Shock, Operating Condition. - The control panel, self-mounted as specified in 4.5.2.2.1, shall be capable of withstanding shock tests as specified in 4.5.2.2.2 and thereafter shall show no damage and meet all performance requirements and product characteristics of this specification.

3.4.3 Temperature and Humidity. - The control panel shall be capable of withstanding the temperature and humidity conditions specified in 4.5.4 and 4.5.5 and thereafter shall show no corrosion or damage and meet the performance requirements and product characteristics of this specification.

3.4.4 Tropical Environment. - The control panel shall be capable of withstanding exposure to the tropical environment specified in 4.5.6 and thereafter shall show no fungus growth, corrosion, or other damage and meet the performance requirements and product characteristics of this specification.

3.4.5 Marine Atmosphere Environment. - The control panel shall be capable of withstanding marine atmosphere environmental conditions specified in 4.5.7 and thereafter shall show no evidence of corrosion or damage and meet the performance requirements and product characteristics of this specification.

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3.5 Clean-up. - Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.6 Workmanship. - The control panel, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges.

3.6.1 The workmanship exhibited in the pre-production sample will be evaluated to determine acceptability. The approved standards of workmanship will thereby become a minimum production requirement for all units offered for acceptance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to ensure that the material to which this specification applies will meet the requirements shown on the applicable drawings. If so stipulated in the contract, order or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required in the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

4.1.2 Pre-production Sample. - Unless otherwise specified for each contract or order, a representative sample of two (2) control panels shall be manufactured using the methods and procedures proposed for the production lot. These shall be submitted for pre-production tests at an activity designated in the contract or by the Bureau of Naval Weapons

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for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on pre-production samples shall be at the contractor's risk. Pre-production samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

4.1.3 Periodic Production Samples. - The tests of 4.4 and 4.5 shall be performed in steps as specified below. Periodic production samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

Step 1. - Upon initiation of production, select one (1) unit from each group of ten (10) units for submission to an activity designated by the procuring agency.

Step 2. - Upon satisfactory compliance of each unit from three (3) consecutive groups, select one (1) unit from each group of twenty-five (25).

Step 3. - Upon satisfactory compliance of each unit from three (3) consecutive groups, select one (1) unit from each group of fifty (50) units. Continue this procedure until a defective unit is found.

Step 4. - When a defective unit is found in any group, the group represented by the sample shall be rejected. Shift to step 1 and proceed as indicated. Disposition of groups represented by failing samples shall be made by the Government.

4.2 Classification of Tests. - The inspection and testing of the control panel shall be classified as follows:

4.2.1 Acceptance Tests. - These tests are to be accomplished on each control panel being submitted for acceptance under contract. These tests are defined in 4.4.

4.2.2 Pre-production and Periodic Production Tests. - Pre-production and periodic production tests are those which are accomplished on samples selected as specified in 4.1.2 and 4.1.3 which are representative of the production of the control panel after the award of the contract to determine that the production meets the requirements of this specification. These are detailed in 4.5. Failure of any sample in any of these tests or requirements will result in the rejection of the lot.

4.3 Test Equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification:

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4.3.1 Oscilloscope with a high persistance screen capable of a 100 millisecond sweep time. (Dumont Model 304A or equivalent)

4.3.2 Test Circuit (see QAP 609).

4.3.3 High-Potential Test Instrument. - An insulation tester rated at 115-volts, 60-cycles, 600-volt-amperes. The tester shall be capable of delivering potentials of 0 to 2500 volts. Means shall be provided for varying the test voltage from minimum to maximum value. An indicator shall be provided for indicating presence of a short circuit in the circuit under test.

4.3.4 Variac. - To be used in conjunction with High-Potential Test Instrument (4.3.3). Must be capable of varying voltage from 0 to 115 V and rated at 600 volt-amperes.

4.3.5 Insulation-Resistance Meter. - A meter capable of measuring insulation resistance of at least three (3) megohms at a potential of 500-volts DC.

4.4 Acceptance Tests.

4.4.1 The contractor shall conduct acceptance tests as specified below to assure that the control panel is in compliance with the requirements of 3.2 and 3.3 (except 3.3.5 and 3.3.8), 3.5 and 3.6.

4.4.2 Test Conditions. - Unless otherwise specified, the control panel shall be subjected to acceptance tests under the following conditions:

4.4.2.1 Temperature: Room ambient 65 degrees Fahrenheit to 95 degrees Fahrenheit (18.33 degrees Centigrade to 35 degrees Centigrade).

4.4.2.2 Altitude: Normal ground

4.4.2.3 Vibration: None

4.4.2.4 Humidity: Room ambient to 95 percent relative maximum.

4.4.3 Test and Inspection Equipment and Facilities: The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity, and when definite requirements are not specified they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.4 Test Procedure.

4.4.4.1 Visual. - The visual tests shall be performed in accordance with NAVWEPS QAP 609 (Visual Tests).

4.4.4.2 Continuity. - Continuity tests shall be performed in accordance with NAVWEPS QAP 609 (Continuity Tests).

4.4.4.3 Standby-Firing Circuit. - The fire circuit shall be tested according to NAVWEPS QAP 609 (Standby-Firing Circuit Test) to determine compliance with the requirements specified in 3.3.2.

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4.4.4.4 Time Delay. - The time delay circuitry shall be tested according to NAVWEPS QAP 609 (Time Delay Test) to determine compliance with the requirements specified in 3.3.2.1.

4.4.4.5 Circuit Breakers. - The circuit breakers shall be tested in accordance with NAVWEPS QAP 609 (Circuit Breaker Test) to determine compliance with the requirements specified in 3.3.3.

4.4.4.6 Operational Tests. - The ISD and FLOOR/GYRO selection circuitry and the insert circuitry shall be tested in accordance with NAVWEPS QAP 609 (Operational Tests) and (Circuit Fault Test) to determine compliance with the requirements specified in 3.3.1.

4.4.4.7 Power Supply. - The Control Panel Mk 264 Mod 1 shall be tested for satisfactory operation under the power limitations established in 3.3.5. This test may be run during the tests given in 4.4.4.6.

4.4.4.8 Dielectric-Insulation Resistance. - Dielectric and insulation resistance tests shall be conducted in accordance with NAVWEPS QAP 609 (Dielectric and Insulation Resistance Tests) to determine compliance with the requirements specified in 3.3.6 and 3.3.7.

4.4.4.9 Preservation, Packing, and Packaging. - Preservation, packing, and packaging of the control panel shall be inspected for conformance to section 5 of this specification.

4.5 Pre-production and Periodic Production Tests.

4.5.1 Vibration

4.5.1.1 Vibration, Transportation, Storage and Handling. - When specified by the procuring agency, the control panel packed and packaged as defined in 5.1 and 5.2, shall be subjected to the following test.

4.5.1.1.1 Mounting. - The control panel, packed and packaged, shall be secured to the mounting bracket of the vibration testing machine in such a manner that the direction of vibration will be along each of the principle axes separately. The equipment may be reoriented on the horizontal vibrating machine between each test, if necessary, to obtain such motion. At no time shall the equipment be mounted in any other way than its normal position.

4.5.1.1.2 Test Procedure. - The package shall be vibrated in each of the test axes as follows:

Frequency Range	Increments of Frequency	(Double) Amplitude
0-12 cps	1 cps	.125 ± .010 in.
13-25 cps	2 cps	.062 ± .008 in.
26-55 cps	3 cps	.031 ± .006 in.

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The package shall be held at resonant frequencies for a minimum of five minutes and shall be held at frequencies 1/2 cycle above and below the resonant frequencies for a minimum of one minute. At the conclusion of the test the control panel shall meet the performance and other requirements specified in this specification.

4.5.1.2 Vibration, Operating Condition.

4.5.1.2.1 Mounting. - The control panel shall be mounted on the vibration testing machine to simulate as realistically as possible the service installation including vibration isolators or mounting devices. The control panel shall be secured to the vibration testing machine in such a manner that the direction of vibration will be along each of the principle axes separately. The control panel may be reoriented on the horizontal vibrating machine between each test, if necessary, to obtain such motion. At no time shall the equipment be mounted in any other way than its normal position.

4.5.1.2.2 Test Procedures. - The control panel shall be energized to perform its normal function and subjected to the following tests. If it is found necessary to repair any damage severe enough to constitute a failure of the equipment the entire test shall be repeated, unless otherwise directed by the Bureau of Naval Weapons or agency concerned.

4.5.1.2.2.1 Exploratory Vibration Test. - To determine the presence of resonances in the control panel, it shall be vibrated at frequencies from 5 cycles per second to 33 cycles per second, at a table vibratory amplitude of 0.010 plus or minus 0.002 inch. The change in frequency shall be made in discrete frequency intervals of 1 cycle per second and maintained at each interval for 15 seconds. The frequencies at which resonances occur shall be noted.

4.5.1.2.2.2 Variable Frequency Test. - The control panel shall be vibrated from 5 cycles per second to 33 cycles per second, in discrete frequency intervals of 1 cycle per second, at the amplitude listed below. At each integral frequency, the vibration shall be maintained for 5 minutes.

Frequency Range (Cycles per sec.)	Table Amplitude (Inch)
5 to 15	0.030 ± 0.006
16 to 25	0.020 ± 0.004
26 to 33	0.010 ± 0.002

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4.5.1.2.2.3 Endurance Test. - The control panel shall be vibrated for a total period of at least 2 hours, at the resonant frequencies chosen by the test engineer. If no resonance was observed, this test shall be performed at 33 cycles per second. The amplitudes of vibration shall be as follows:

Frequency Range (Cycles per sec.)	Table Amplitude (Inch)
5 to 15	0.030 ± 0.006
16 to 25	0.020 ± 0.004
26 to 33	0.010 ± 0.002

At the conclusion of the tests specified by 4.5.1.2.2.3, the control panel shall meet the performance and other requirements specified in the specification.

4.5.2 Shock

4.5.2.1 Shock, Transportation, Storage, and Handling. - When specified by the procuring agency, the control panel, packed and packaged as defined in 5.1 and 5.2, shall be subjected to the following rough handling and exposure tests.

4.5.2.1.1 Free Fall Drop Test. - The container shall be dropped cornerwise from a height of 30 inches onto a steel, concrete or stone surface of sufficient mass to absorb the shock without deflection in such a manner that the corner of the container absorbs the full force of the fall. This test shall be repeated until each of the eight corners of the container has received a fall. (The height of 30 inches refers to the distance from the steel, concrete or stone surface to the nearest corner of the container when suspended prior to the fall.) The fall shall be a free fall, in that no ropes or other suspending media are attached to the container during the fall.

4.5.2.1.2 Cyclic Exposure Test. - At the conclusion of the free fall drop test, the package shall stand undisturbed for 24 hours and then subjected to the cyclic exposure test as follows:

Overnight at 120 degrees to 130 degrees Fahrenheit
Two hours of water spray at 50 degrees to 60 degrees Fahrenheit
Two hours at minus 10 degrees to 0 degrees Fahrenheit
Two hours of water spray at 120 degrees to 130 degrees Fahrenheit
Two hours of water spray at 50 degrees to 60 degrees Fahrenheit
Overnight at 35 degrees to 50 degrees Fahrenheit
Four hours at 120 degrees to 130 degrees Fahrenheit

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Two hours of water spray at 50 degrees to 60 degrees Fahrenheit
Two hours at 35 degrees to 50 degrees Fahrenheit
Overnight at 120 degrees to 130 degrees Fahrenheit
Two hours of water spray at 50 degrees to 60 degrees Fahrenheit
Two hours at minus 10 degrees to 0 degrees Fahrenheit
Three hours at 35 degrees to 50 degrees Fahrenheit
Overnight at 120 degrees to 130 degrees Fahrenheit

4.5.2.1.3 At the conclusion of the free fall drop and cyclic exposure test, all materials and components comprising the method of preservation shall be free from damage or evidence of displacement which affect the utility of the method of preservation. The control panel, when examined, shall show no visible signs of damage or evidence of corrosion and shall meet the performance and other requirements specified in this specification.

4.5.2.2 Shock - Operating Condition.

4.5.2.2.1 Mounting. - The control panel shall be so mounted on the mounting bracket of the test machine as to simulate as realistically as possible the service installation. The control panel shall be secured to the mounting bracket of the test machine in the same manner that it will be secured on shipboard and in such a manner that the shock blows will be parallel to each of the three principle axes.

4.5.2.2.2 Test Procedure. - The control panel shall be energized to perform its normal function and subjected to the following test. If it is found necessary to repair any damage severe enough to constitute a failure of the equipment, the entire test shall be repeated, unless otherwise directed by the Bureau of Naval Weapons or agency concerned.

4.5.2.2.2.1 Shock Test. - The shock test shall consist of 9 blows applied in 3 groups of 3 blows each with each group of blows being applied parallel to each of the three principle axes of the control panel. Each group of 3 blows shall consist of 9 blows where the height of the hammer drop shall be 12 inches, 1 blow where the height of the hammer drop shall be 24 inches, and 1 blow where the height of the hammer drop shall be 36 inches. At the conclusion of this test, the control panel shall meet the performance and other requirements specified in this specification.

4.5.3 Radio Frequency Conducted Interference. - The control panel shall be tested in accordance with Specification MIL-I-16910 (Undesired Radiation Test) to determine compliance with the requirements specified in 3.3.5.

4.5.4 Temperature Test. - The control panel shall first be placed within a test chamber with power off and all covers in place. The ambient air temperature within the test chamber shall then be raised to

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80 degrees Fahrenheit with relative humidity between ninety-five percent (95%) and ninety-eight percent (98%). During the raise in temperature to 80 degrees Fahrenheit, the control panel shall have its covers in place and power on. The ambient air temperature rise shall not exceed 10 degrees per hour with a maximum of 30 degrees in 8 hours. The control panel shall operate satisfactorily at any time during this changing of the ambient air temperature. After 10 hours at 80 degrees Fahrenheit, the control panel shall then meet the performance and other requirements specified in this specification for a period of 50 hours under the conditions described. Deterioration or corrosion of any internal or external components which could in any manner prevent the equipment from meeting operational requirements during service life shall provide reason to consider the equipment as having failed to meet the test.

4.5.5 Humidity Tests

4.5.5.1 Procedure I. - The control panel shall be placed in a test chamber and set up to simulate installed conditions. The test chamber shall be vented to the atmosphere to prevent buildup of pressure. Prior to the starting of the test period, the chamber temperature shall be between 68 degrees and 100 degrees Fahrenheit. Distilled or demineralized water having a pH value of between 6.5 and 7.5 at 25 degrees Centigrade (77 degrees Fahrenheit) shall be used to obtain the desired humidity. The temperatures in the chamber shall be gradually raised from between 20 degrees and 38 degrees Centigrade (68 degrees to 100.4 degrees Fahrenheit) to 71 degrees Centigrade (160 degrees Fahrenheit) during a 2-hour period. The temperature of 71 degrees Centigrade (160 degrees Fahrenheit) and a relative humidity of 95 percent shall be maintained during the next 6-hour period. During the following 16-hour period, the temperature in the chamber must drop at a uniform rate to 20 degrees to 38 degrees Centigrade (68 degrees to 100.4 degrees Fahrenheit) which constitutes one cycle. The relative humidity throughout the cycle shall be 95 percent. The cycle shall be repeated a sufficient number of times to extend the total time of the test to 240 hours (10 cycles). Within one hour after the conclusion of the 240-hour period, the control panel shall be inspected for evidence of deterioration or corrosion and shall then meet the performance and other requirements specified in this specification.

4.5.5.2 Procedure II. - The control panel shall be placed in a chamber having a temperature of 49 degrees Centigrade (120 degrees Fahrenheit) and humidity of 95 percent. The test conditions shall be maintained for 360 hours. At the conclusion of this period, the control panel shall be operated within one hour and shall meet the performance and other requirements specified in this specification.

4.5.6 Tropical Environment. - The control panel shall be subjected to the following tropical environmental test.

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4.5.6.1 Test Procedure. - The control panel shall be placed in a mold chamber maintaining an internal temperature of 30 plus or minus 2 degrees Centigrade (86 plus or minus 3.6 degrees Fahrenheit) and a relative humidity of 95 plus or minus 5 percent and sprayed with the suspension of mixed spores. These conditions shall be maintained for 28 days. At the conclusion of the test, the control panel shall be allowed to return to room temperature before performance checking. A list of four groups of fungi are listed below. One type of fungi from each group shall be used. In preparing the spore suspension, distilled water (having a pH value between 5.8 and 7.2 at a temperature between 22 degrees and 32 degrees Centigrade) (72 degrees and 89 degrees Fahrenheit) shall be prepared by sterilization in convenient containers (approximately 100 milliliters each). Approximately 10 milliliters of the solution shall then be introduced directly into a tube culture of one fungus and shaken vigorously in order that a well sporulated suspension will result without disturbing the agar. This process shall be repeated for each type of fungus. The separate spore suspensions from the types of fungi shall then be mixed together to provide a composite suspension. Actively growing cultures between 7 to 21 days old after initial innoculation shall be used in the preparation of the spore suspension. After preparation, the spore suspension shall not be kept for more than a 24-hour period at temperatures from 2 degrees Centigrade to 7 degrees Centigrade (35 degrees Fahrenheit to 45 degrees Fahrenheit).

GROUP I	Chaetomium globosum - 6205 or Myrothecium verrucaria 9095.
GROUP II	Memononiella echinata 9597 or Aspergillus niger - 6275
GROUP III	Aspergillus flavus - 10836 or Aspergillus terreus - 10690
GROUP IV	Penicillium ochrochloron 9112 Penicillium citrinum - 9849

4.5.7 Marine Atmosphere Environment. - When specified by the procuring agency, the control panel shall be subject to the following sea atmosphere conditions.

4.5.7.1 The control panel assembled, sealed, and ready for service use, shall be mounted in the exposure chamber maintained at 35 degrees Centigrade (95 degrees Fahrenheit). A salt solution (see 4.5.7.1.2) spray creating a finely divided, wet dense fog shall be introduced into the chamber. The conditions maintained in all parts of the exposure zone shall be such that a suitable receptacle placed at any point in the exposure zone will collect from 0.5 to 3 ml of solution per hour for each 80 square cm of horizontal collecting area (10 cm diameter) based on an average of at least 16 hours. The length of the marine stmosphere test shall be not less than 50 hours.

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4.5.7.1.1 Salt Solution. - The salt used shall be sodium chloride containing on the dry basis not more than 0.1 percent of sodium iodide and not more than 0.2 percent of total impurities. The solution shall be prepared by dissolving 20 plus or minus 2 parts by weight of salt in 80 parts by weight of distilled or other water containing not more than 200 parts per million of total solids. The solution shall be adjusted to and maintained at a specific gravity of from 1.126 to 1.157 and at a pH of between 6.5 to 7.2 when measured at a temperature between 33 degrees and 36 degrees Centigrade (92 degrees and 97 degrees Fahrenheit). Only C. P. hydrochloric acid or C. P. sodium hydroxide shall be used to adjust the pH.

4.5.7.2 At the conclusion of the marine atmosphere test, the control panel shall meet the performance and other requirements specified in this specification. Salt deposits resulting from the test may be removed prior to operation.

4.5.8 Operating Life. - The control panel shall be subject to the following accelerated life test:

4.5.8.1 Conditioning. - In order to establish a reference for the measurement of operating parameters and a valid basis for comparison of test results, the control panel shall be subjected to the following conditioning process.

4.5.8.1.1 Normal Operation. - The control panel shall be set up in a room in an ambient temperature of 25 degrees plus or minus 5 degrees Centigrade (77 degrees plus or minus 9 degrees Fahrenheit) and a 50 plus or minus 3 percent relative humidity and operated under conditions of nominal line power voltage, frequency input, and output load conditions, in accordance with the requirements of this specification. Following an adequate warm-up period, critical performance parameters as specified in this specification, plus any special tests deemed necessary by the Government inspector, shall be measured and logged as "reference test data" for comparison with subsequent tests.

4.5.8.1.2 High Temperature Operation. - The control panel shall then be set up in a room which shall be maintained at a temperature of 50 degrees plus 5 degrees minus 0 degrees Centigrade (122 degrees plus 9 degrees minus 0 degrees Fahrenheit). The room shall be maintained at not less than 90 percent relative humidity. The control panel shall be energized and operated under these conditions at nominal line power voltage, frequency input, and output load conditions, for a period of 2 hours. Near the end of this period and preferably during the last half hour, performance test data shall be taken for comparison with the reference test data. Minor adjustments to operating controls will be permitted during this 2-hour test run to insure optimum equipment performance.

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4.5.8.2 Test Cycling. - Without shutting down the equipment and immediately following the 2-hour high temperature conditioning run, the equipment shall be subjected to a series of test cycles while operating in a steady state ambient temperature of 50 degrees plus 5 degrees minus 0 degrees Centigrade (122 degrees plus 9 degrees minus 0 degrees Fahrenheit). The relative humidity shall be maintained at not less than 90 percent.

4.5.8.2.1 Test cycling shall be as follows: Increase input power voltage to 110 percent nominal and operate in this condition for a period of 1 hour and 15 minutes; then decrease voltage to 90 percent nominal and operate in this condition for 1 hour; at the end of this period increase input voltage to nominal and operate for 1/2 hour, at the end of which time the equipment shall be de-energized for 15 minutes. During the 1/2-hour period of operation at nominal line input, repeat measurement of operating parameters. At the end of the 15 minute shutdown period, energize the control panel and immediately decrease the input voltage to 90 percent nominal; operate in this condition for 1 hour; increase input voltage to 110 percent nominal and operate in this condition for 1 hour and 15 minutes; decrease voltage to nominal and operate for a period of 1/2 hour, again repeating measurements of operating parameters, followed by a 15 minute shutdown of the control panel.

4.5.8.2.2 The test cycle specified in 4.5.8.2.1 shall be repeated without interruption for a period of 360 hours (15 complete days).

4.5.8.3 Procedure In Case of Shutdown. - Should a single shutdown for more than 1 hour or a total shutdown of 8 hours during the 15 days of continuous operation be necessary, the continuation of the test shall be for 15 full days from the time of resumption of the run.

4.5.8.4 Test Data. - Test data accumulated during the accelerated life tests, including details of all failures, shall be provided and included in the report on acceptability tests.

4.6 Failure of the control panel to meet any of the requirements and tests of this specification shall be considered cause for rejection.

4.7 Waiver of Tests. - Tests specified in 4.5.1.1 and 4.5.2.1 may be waived by the procuring agency for level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.8 Functional Tests. - At the conclusion of the tests specified in 4.5.1.1 and 4.5.2.1, functional and other tests, as applicable, shall be conducted on the control panel to determine freedom from operational malfunction and compliance with all the requirements of section 3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A

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5.1.1.1 Cleaning. - The exterior surfaces only of the control panel shall be cleaned in accordance with process C-1 of Specification MIL-P-116.

5.1.1.2 Drying. - The exterior surfaces only of the control panel shall be dried in accordance with procedure D-1 or D-4 of Specification MIL-P-116.

5.1.1.3 Preservative Application. - None required.

5.1.1.4 Unit Packaging. - The control panel shall be individually packaged in accordance with Method IIa of Specification MIL-P-116. Movable parts, susceptible to damage, shall be cushioned, secured, blocked and braced to prevent movement prior to packaging. Material for gaskets used to fabricate the package shall conform to Specification MIL-G-12803, Type 1, Class 1, Group 2, Grade 1, .125 inch thick. The adhesive for cementing the gasket shall conform to Specification MIL-A-5092, Type III. The humidity indicator for the package shall conform to Specification MIL-I-26860, Type II.

5.1.2 Level B. - Not applicable.

5.1.3 Level C.

5.1.3.1 Cleaning, Drying and Preservative Application. - The cleaning, drying and preservative application of the control panel shall be in accordance with para 5.1.

5.1.3.2 Unit Packaging. - The control panel shall be individually packaged in accordance with the manufacturer's commercial practice, when such practices have been proven by test to conform to all applicable requirements. The packaging shall be adequate to provide protection from physical damage, corrosion or other forms of deterioration during direct shipment from supply source to the first receiving activity for immediate use.

5.2 Packing.

5.2.1 Level A. - The control panel, packaged as specified in para 5.1, shall be individually packed in (overseas type) wood cleated plywood or nailed wood box conforming to Specification PPP-B-601, Style A or B, or PPP-B-621, Class 2, Style 2 or 4, respectively. The control panel shall be cushioned, blocked and braced to provide protection from damage during transportation, handling and storage. The box shall be provided with an inspection port for viewing the indicator. The containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

5.2.2 Level B. - The packing shall be the same as for Level A, except the shipping containers shall conform to the domestic type.

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5.2.3 Level C. - The control panel, packaged as specified in para 5.1, shall be individually packed in a manner to insure carrier acceptance for safe delivery at destination, at the lowest rate. Containers shall be in accordance with the Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 Normal Marking. - In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Military Standard MIL-STD-129, including the serial number of the control panel contained therein.

5.3.2 Special Marking. - The following special marking shall appear on the top, one (1) side and one (1) end of each shipping container.

DELICATE INSTRUMENT
HANDLE WITH CARE

6. NOTES

6.1 Intended Use. - Control Panel Mk 264 Mod 1 is intended for use aboard vessels having Torpedo Tubes Mark 32 Mods 0, 1, and 2.

6.2 Ordering Data. - Procurement documents should specify the specific title, number, and date of this specification, exceptions to this specification, applicable drawings, and other documents, and whether pre-production samples are required (see 3.2.1). The selection of applicable levels of preservation and packaging and packing of Section 5 shall abide. Selection of Level C shall not be made unless immediate use is definitely known.

6.2.1 Criteria for use of proper level of preservation, packaging and packing shall be as follows:

For Level A. - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B. - This level shall be used only when it is definitely known that the item will be held in covered storage, overseas locations, for 6 months or less, or in domestic locations for an indefinite period.

For Level C. - This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

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